

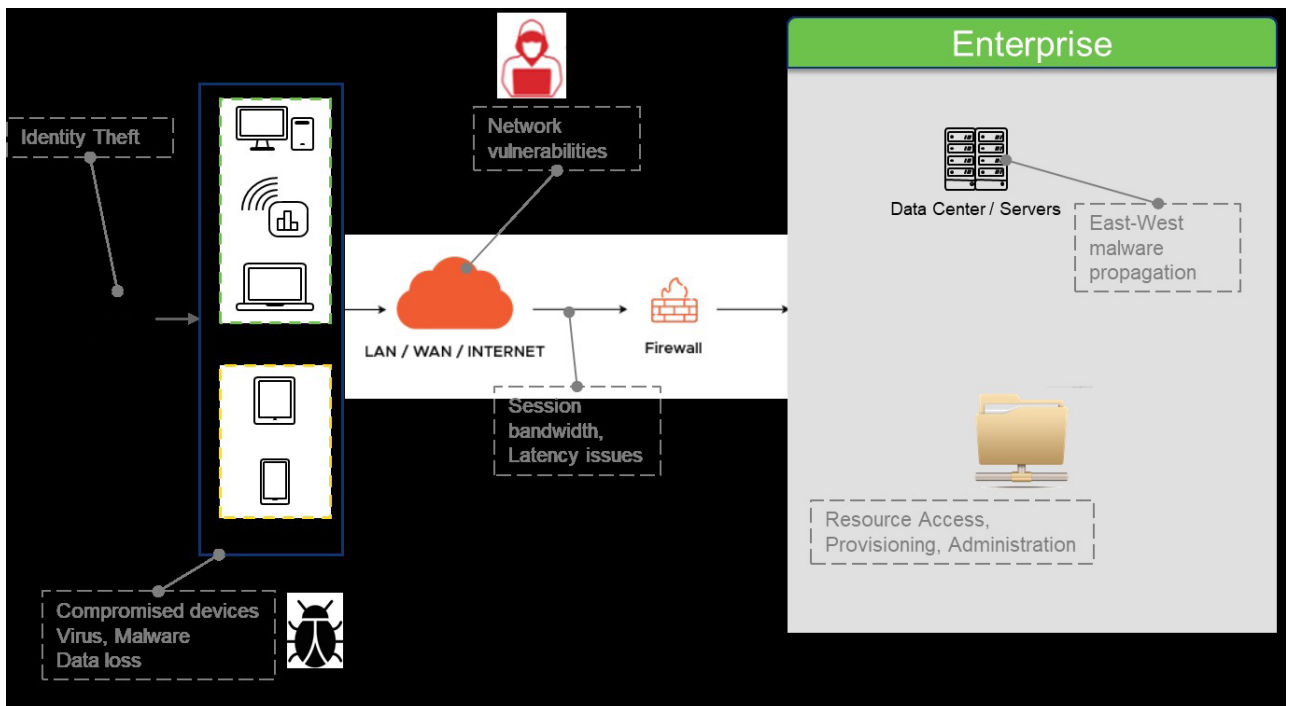


Figure 1: Threats and issues with remote work and BYOD

there, it propagates laterally in an east west direction to take control over more resources. Data gets transferred to the end point and can get into the wrong hands. User sessions over the network are prone to hacking.

How VDI handles security vulnerabilities

VDI is built from the ground up to address the various threats mentioned just now. Figure 2 summarises the

controls imposed at each stage by VDI.

VDI leverages proprietary protocols for session streaming to the end point. Popular protocols in use are ICA from Citrix, and PCoIP and Blast from VMware. User sessions are encrypted end-to-end so that hackers cannot tamper with them during transmission.

Multiple levels of authentication are enforced for better security and to protect identity hijacks. Tokens, biometrics, and certificates over another

communication channel are leveraged as second factor authentication to ensure only the right user gets access.

No data gets transferred to or from the end user device. Posture checks and conditional access allow deployment of security policies. VDI vendors have amplified security controls with anti-keyloggers and screen capture disablers to protect enterprise data and applications.

Even though network segmentation is not an integral part of VDI, it complements the VDI security posture in data centres to control east west traffic. This is important, as the adversary who somehow manages to pass through enterprise doors tries to move laterally to gain control over more and more devices and services in the enterprise. Network segmentation imposes security checks at each component, which stops unwanted east west traffic.

Conditional access

Conditional access is a very important element in the VDI security posture. It includes a contextual policy engine that runs posture checks on end points and

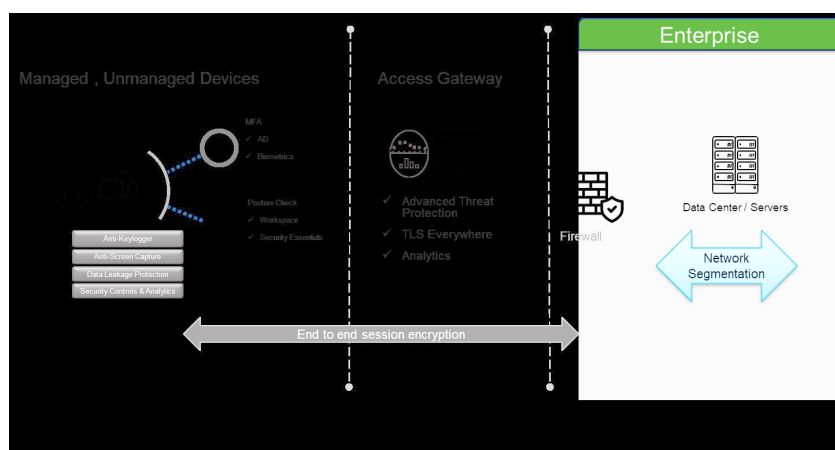


Figure 2: Security measures in a VDI landscape